

GAME DESIGN DOCUMENT (GDD)

NUMERIA ONLINE — *Where Calculus Is the Blade*

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1. High Concept

1.1 Overview

Field	Detail
Working title	NUMERIA ONLINE
Genre	Math-powered MMORPG (Combat + Puzzle + Progression)
Perspective	Third-person, stylized low-poly fantasy
Platforms	PC (primary), Mac, tablet (companion client)
Business model	Free-to-play, cosmetic + optional convenience, no pay-to-win
Session length	15–45 min core loop, 60–90 min raid
Target audience	Core MMO players (16–40), lapsed learners, casual-puzzle crossover, cooperative/social players

1.2 One-Sentence Pitch

An MMORPG where your real mental-math skill is the source of your character's magical power — solve fast arithmetic under pressure to strike world bosses, and prove mastery of each mathematical discipline to unlock the epic gear and forbidden zones beyond it.

1.3 Vision Statement

NUMERIA makes mathematical cognition feel heroic. Every problem solved is a spell cast, a blow landed, a seal broken. The world is literally structured by mathematics, so growing as a player and growing as a mathematician are the same act. The design goal is **mastery without homework fatigue**: the game must feel like a combat fantasy first and an educational experience second, never the reverse.

1.4 Player Fantasy

"I am a raid mage whose real mental speed powers my spells — the sharper my mind, the harder I hit, and the more of reality I can perceive and command."

A note on terminology: throughout this document, "mental calculus" and "mental math" refer to **mental calculation and arithmetic fluency** — the fast, in-the-head evaluation of expressions — which is the core combat verb of the game. Formal calculus (derivatives, integrals, limits) appears only as late-game world logic and boss mechanics (Tier 8+), never as split-second combat input. This preserves accessibility while honoring the title.

1.5 Design Pillars

1. **Math is magic.** Every player action is grounded in a calculation. The fantasy is that *your mind* is the weapon — not a stat sheet.
2. **Skill-based progression.** Power comes from demonstrated competence, not grind time alone. A Tier-5 player is genuinely better at algebra than a Tier-4 player.
3. **Readable under pressure.** Challenges are fast but always legible. Failure is informative, never punishing or humiliating.
4. **Cooperation over gatekeeping.** The MMO layer lets players of different tiers contribute together. No one is dead weight because they're weak at one operation.
5. **Mastery, not worksheets.** Every interaction is anchored to combat fantasy, juicy feedback, cooperative stakes, and tangible rewards. The moment it feels like homework, the design has failed.

2. Setting & Lore

2.1 The World of Numeria

Numeria is a realm whose physical laws are not fixed constants but **mathematical resonances** — living equations that sustain reality. The land is divided into **Dominions**, each governed by a foundational mathematical discipline. Reality in each Dominion is structured by progressively more complex mathematics, so the "aether" of a region literally cannot be perceived, let alone manipulated, by a mind that has not internalized the underlying discipline.

This is the in-fiction reason for zone gating: **a player cannot enter a Dominion until their cognitive resonance matches its frequency.** A mind that has not mastered fractions cannot perceive the Fractal Forest; to them it appears as an impenetrable static fog. Mastery is a key of cognition, not an arbitrary level requirement.

2.2 The Source of Power

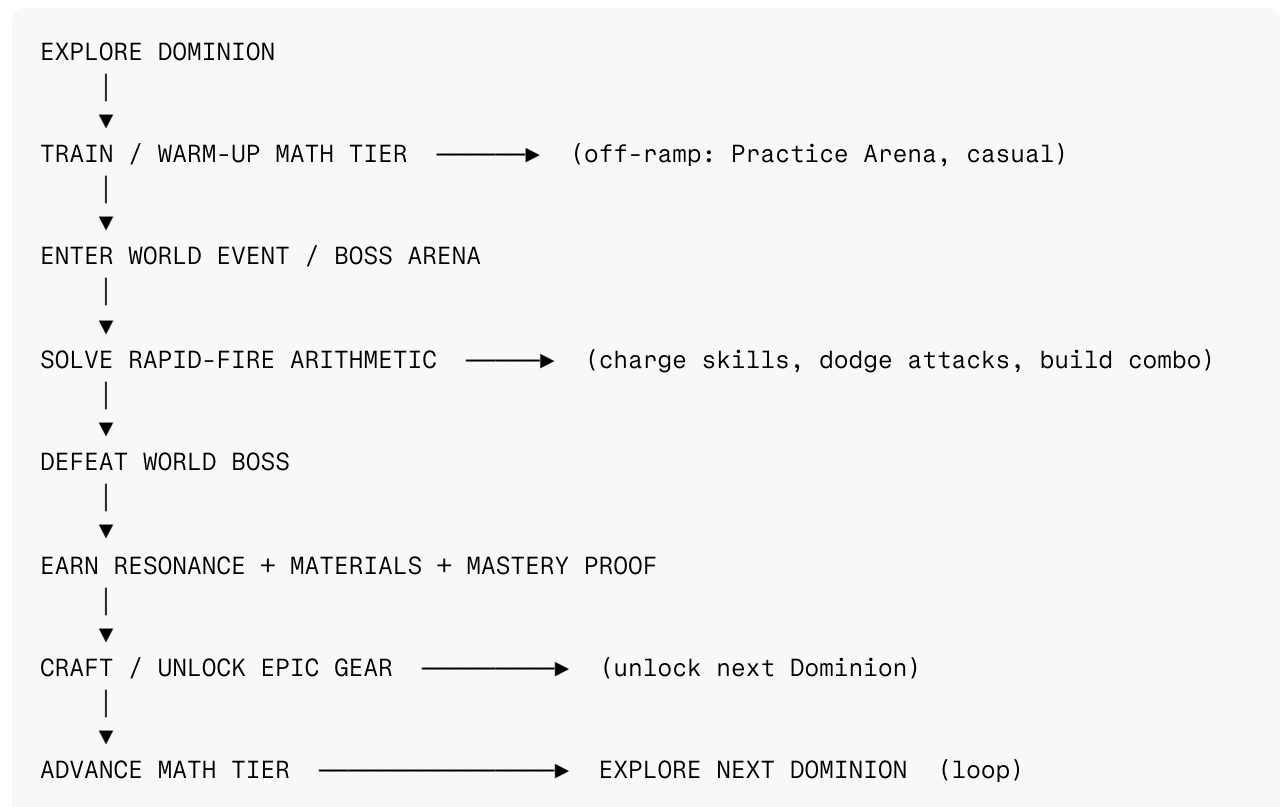
In Numeria, magic is computation. To cast a spell, a sorcerer must *evaluate an expression in real time* — the speed and accuracy of their mental arithmetic determines the spell's power. Thus the most powerful adventurers are not those with the heaviest armor but those with the sharpest minds. Epic gear is not looted at random; it is **forged from proofs** — artifacts that can only be wielded by a mind capable of understanding the mathematics encoded in them.

2.3 The World Bosses — The Discordant

Reality is under assault by **The Discordant**: ancient entities of incoherent mathematics that seek to unravel the equations sustaining Numeria. Each world boss is an embodiment of a corrupted mathematical concept — a Prime Golem whose body is factorization, a Percentile Wraith that distorts probabilities, a Calculus Leviathan that bends continuity itself. Defeating them requires not just power but precision: you must *correct the math* to dispel their corruption.

3. Core Gameplay Loop

3.1 The Loop



3.2 The 90-Second Boss Phase (Worked Example)

A concrete feel for combat. Boss: **The Percentile Wraith**, Tier 3.

- **0:00–0:10 — Telegraph.** Boss charges "Probability Collapse." A large expression appears on screen: *What is 15% of 240?* Timer bar fills (4.5s for this tier).

- **0:10 — Resolve.** Correct answer (36) lands a **Resonance Bolt** for damage and adds a combo counter. Wrong/timeout → boss lands a hit, player staggered, combo resets.
- **0:15–0:30 — Chain.** Three quick-fire problems appear in succession (mentally chaining operations). Each correct answer adds to a damage multiplier. A 5-streak triggers **Overdrive**, doubling the next bolt.
- **0:30 — Boss mechanic.** "Shrouded Field": four problems appear, only one is "true" (others are decoys with false logic). Player must identify the correct one — pattern recognition, not just speed.
- **0:45–1:15 — Burn phase.** Boss enrage timer; problems come faster and harder (within the tier). Streaks determine final damage. The team coordinates: Tank-player soaks hits by solving shield problems, DPS-players solve attack problems, Support solves heal/ cleanse problems.
- **1:15 — Finisher.** A final "proof" puzzle: solve a short multi-step expression. Big damage, cinematic takedown.

Total problems solved in 90s: ~18–25. Each is fast, legible, and visually tied to a combat action. The player *feels* like a battle-mathematician, not a student doing drills.

4. Mental Calculus Combat System

4.1 How Problems Map to Combat

Player action	Math interaction	Combat result
Basic attack	Solve a simple expression	Damage scaled by speed + streak
Charged attack	Solve a multi-step expression	Heavy damage, longer timer
Defend / shield	Solve a problem within a tighter window	Absorb incoming boss hit
Dodge	Quick mental check (true/false or estimate)	Evade telegraphed attack
Heal / cleanse	Solve a problem of the support type	Restore party HP / remove debuff
Ultimate	Solve a "proof" (chained expression)	Massive AoE damage

4.2 Answer Input UX

- **Primary input:** number pad / keyboard for desktop, large numeric buttons for tablet.
- **Multiple choice** for harder tiers (reduce motor cost, keep cognitive cost).
- **Voice / gesture** optional for accessibility (see §10).
- Problems render large and high-contrast; never obscured by effects.
- A subtle "thinking pulse" visualization shows remaining time without anxiety-inducing countdown beeps (toggleable).

4.3 Failure Model — Rewarding, Not Punishing

This is critical to avoiding worksheet fatigue. Failure is **never instant death or humiliation**:

- **Wrong answer** → reduced damage, not zero. You still participate.
- **Timeout** → boss lands a hit, but player regains footing. Streak resets but play continues.
- **Streak forgiveness**: first wrong answer in a streak halves the combo rather than wiping it.
- **No public failure**. Other players see your *results* (damage numbers, support output), never your *wrong answers*.
- **Assist mode** (opt-in): extends timers, offers multiple-choice, reduces problem complexity by one sub-tier. Clearly labeled, no stigma — see §10.

4.4 Combo & Resonance System

- **Streak multiplier**: consecutive correct answers build a damage multiplier (1x → 2x → 3x cap).
- **Resonance meter**: a party-wide resource filled by correct answers; fuels group ultimates and raid-wide buffs.
- **Overdrive**: triggered at high streaks — slows the timer slightly and amplifies the next answer, rewarding hot streaks without making them mandatory.

4.5 Adaptive Difficulty

The engine continuously estimates each player's **response time and accuracy** per operation type and dynamically adjusts:

- **Band within a tier**: if a player is struggling, problems shift to the lower complexity band of the tier (e.g., 2-digit instead of 3-digit multiplication); if excelling, to the upper band.
- **Targeting weaknesses gently**: the system surfaces a player's weaker operation slightly more often (spaced practice) but never to the point of frustration.
- **This keeps every player in flow** — challenged but not crushed — which is the single most important anti-burnout mechanism in the game.

4.6 Anti-Cheat & Calculator Handling

Since skill is the source of power, integrity matters:

- **Server-authoritative problem generation**; answers validated server-side.
- **Problem randomization** per player per encounter (no shared answers to copy).
- **In-world recognition of legitimate variance** — a player using assist mode is flagged as such for fair matchmaking, not penalized.
- **External calculators are unnecessary by design**: the game *is* the calculator. Using one would be slower than playing.

5. Puzzle System

5.1 Design Contrast

Where combat tests **fluency under pressure**, puzzles test **understanding and pattern recognition** at a slower, more contemplative pace. This contrast is the rhythm of the game — intense boss fights alternate with exploratory puzzle-dungeons.

5.2 Puzzle Types

- **Spatial arithmetic:** arrange tiles so rows/columns sum to targets (magic-square style).
- **Sequence & pattern:** identify the rule governing a sequence of numbers or shapes.
- **Logic gates:** boolean-style puzzles controlling dungeon mechanisms.
- **Proof dungeons:** multi-room challenges where each room solves one step of a larger expression — cooperative, with different players handling different steps.
- **World logic puzzles:** the Dominion's environment itself obeys a rule (e.g., in the Geometric Labyrinth, movement must follow vector rules). Understanding the rule lets you traverse.

5.3 Integration with Combat

Puzzles reward permanent tools and gear modifiers (e.g., a puzzle-dungeon yields a rune that changes a combat operation type or extends a timer). Puzzles thus feed back into combat, so both systems reinforce each other rather than feeling disconnected.

6. Math Tier Progression System

6.1 Design Principle

Moment-to-moment **combat** stays focused on **arithmetic and mental calculation** — fast, readable, satisfying. Advanced topics (algebra, geometry, calculus) appear as **puzzle mechanics, boss modifiers, world logic, and gear crafting proofs**, not as split-second combat inputs. This preserves accessibility: a player need not solve a derivative in 3 seconds, but they must *understand* the derivative to craft the epic gear that the Calculus Dominion offers.

6.2 Tier Table

Tier	Discipline	Example combat challenge	Example puzzle / boss mechanic	Gear rarity unlocked	Dominion unlocked
0	Foundations	$3 + 4$, $9 - 2$	Counting, ordering	Common	The Cradle
1	Integer Arithmetic	12×7 , $144 \div 12$	Multi-step drills	Uncommon	Numeria Prime
2	Fractions & Decimals	$0.5 + 0.25$; $3/4$ of 20	Tile-sum puzzles	Rare	The Fractal Forest
3	Percentages & Ratios	15% of 240; simplify 12:8	Probability puzzles	Rare+	The Bazaar of Percent

Tier	Discipline	Example combat challenge	Example puzzle / boss mechanic	Gear rarity unlocked	Dominion unlocked
4	Pre-Algebra / Order of Ops	$3 + 4 \times 2$; (evaluate)	Expression-gate dungeons	Epic	The Operator's Reach
5	Algebra	Solve $x + 7 = 22$ (combat: estimate root)	Variable-logic puzzles	Epic	The Algebraic Wastes
6	Geometry & Trigonometry	Estimate angles / areas (combat: spatial)	Vector-movement dungeons	Epic+	The Geometric Labyrinth
7	Number Theory	Identify primes; modular sums	Factorization vaults	Legendary	The Prime Vault
8	Calculus	Estimate limits / rates (boss mechanic)	Continuity / rate puzzles	Legendary	The Calculus Citadel
9	Abstract / Applied	Probability distributions, combinatorics (raid mechanics)	Proof-style raid encounters	Mythic	The Hilbert Expanse

6.3 Mastery Proofs

To advance a tier, a player must complete a **Mastery Proof**: a curated sequence of challenges (mix of combat fluency + puzzle understanding) demonstrating competence, not just speed. Mastery is measured by **accuracy and consistency over time**, so a thoughtful player beats a reckless fast one. This is what makes progression feel earned rather than grindy.

6.4 Spaced Practice & Retention

To prevent skill decay, the system uses **spaced retrieval**: previously mastered operations reappear in later zones at low frequency, keeping them fresh without blocking progress. This mirrors evidence-based learning science and keeps old tiers relevant.

7. Gear & Loot System

7.1 Design Principle

Epic gear is **earned through demonstrated mastery**, not dropped at random or bought. More importantly, gear enhances **expression**, not just raw numbers: better gear gives more time, alternate problem types, combo forgiveness, team buffs, and puzzle utility — so a skilled player in lower gear can still outperform a weaker player in higher gear. This keeps skill, not gear, as the primary determinant of power.

7.2 Gear Acquisition

- Defeating world bosses yields **Mastery Materials** themed to their Dominion.
- Materials + a completed **Mastery Proof** → craft the Dominion's epic set.
- Sets are tiered to math tiers; wearing a set requires the corresponding mastery (you cannot wield gear encoding math you don't understand).

7.3 Epic Gear Sets by Tier (Examples)

Set	Tier	Signature effect
<i>Vestments of the Sum</i>	1	+1s on basic attacks; minor combo protection
<i>Mantle of the Fraction</i>	2	Converts some integer problems to fraction variants (more resonance per solve)
<i>Cape of the Percentile</i>	3	Reveal one decoy in "Shrouded Field" boss mechanic
<i>Gauntlets of the Operator</i>	4	Reorder a problem's operations once per fight (strategic tool)
<i>Regalia of the Variable</i>	5	Substitutes a known value into an expression (pre-computed assist)
<i>Aegis of the Geometer</i>	6	Visualizes spatial/angle hints for geometry combat
<i>Crown of Primes</i>	7	Reveals factorization of a boss's shield value
<i>Mantle of the Differential</i>	8	Slows rate-based boss mechanics; extends enrage timers
<i>Panoply of the Abstract</i>	9	Party-wide resonance amplification in raid proofs

7.4 Non-Power Rewards

Core cosmetic progression — skins, titles, mounts, and social flair — is available through all playstyles and is never locked behind math skill, ensuring players who struggle with math still have rich reward paths and never feel second-class. A small set of prestige cosmetics may celebrate demonstrated mastery (e.g., seasonal proof-dungeon rewards), but these are purely cosmetic and carry no power advantage.

8. World Zones & Gating Rationale

8.1 Gating Pattern

Each Dominion gate has both a lore reason and a design reason. The design reason follows a strict pattern: each zone introduces exactly one new cognitive load and one new combat verb. This ensures difficulty curves up smoothly and the player is never overwhelmed by multiple new concepts at once.

8.2 Zone Table

Dominion	Math tier	World boss	Gameplay demand	Why lower tiers cannot enter
The Cradle	0	The Counting Beast	Basic fluency, learn input	Onboarding; no gate
Numeria Prime	1	The Prime Golem	Multi-digit arithmetic + combo	Aether requires integer resonance; static fog blocks the uninitiated
The Fractal Forest	2	The Fraction Mantis	Part/whole reasoning	Trees split into fractions; a mind without fraction-sense sees only disjointed mist

Dominion	Math tier	World boss	Gameplay demand	Why lower tiers cannot enter
The Bazaar of Percent	3	The Percentile Wraith	Probability & ratio under pressure	The market's reality shifts by percentage; perception requires ratio intuition
The Operator's Reach	4	The Bracket Warden	Order-of-operations priority	Sequences of operations chain; missing precedence causes "logic collapse" in the zone
The Algebraic Wastes	5	The Variable Drake	Unknowns & estimation	The wastes are defined by unknowns; without algebra, the landscape is unreadable
The Geometric Labyrinth	6	The Angle Serpent	Spatial reasoning, vectors	Movement obeys geometric law; non-geometric minds cannot navigate
The Prime Vault	7	The Factor Colossus	Divisibility & modular reasoning	The vault's doors open only to those who sense prime structure
The Calculus Citadel	8	The Continuity Leviathan	Rates, limits, change over time	Time flows differently here; only minds grasping change can perceive it
The Hilbert Expanse	9	The Abstract (raid)	Combinatorics, probability distributions	The expanse is pure abstraction; only the mastered mind can hold form here

8.3 The Rationale, Summarized

The gating is **not arbitrary difficulty-walling**. It is the fiction and the mechanics agreeing: each Dominion's reality is *constituted* by a mathematical discipline, so entering it requires the cognitive capacity to process that discipline. This transforms "you must be this tall to ride" into a narratively and mechanically coherent journey of the mind growing to encompass more of reality. The player *feels* their own intellectual expansion as the world literally opens to them.

9. MMO Systems

9.1 Cooperative Math Design

A core anti-burnout and anti-toxicity feature. Encounters are designed so players of **different tiers and different operation-strengths** can contribute meaningfully together:

- **Role diversity:** Tank, DPS, and Support roles solve *different problem types* (shield problems, attack problems, heal problems). A player strong at multiplication but weak at fractions can specialize where they shine.
- **Party scaling:** problems scale in count, not just difficulty, so a larger party shares the cognitive load.
- **Carry-prevention without exclusion:** a weaker player is given high-value-but-achievable problems (e.g., the "true/false decoy identification" role), so they contribute real value and learn by doing, rather than being dead weight or carried.
- **Mentor Mode:** veteran players can join lower-tier content as Mentors; their damage is capped (no trivializing), but they earn mentor currency by helping newcomers succeed.

Incentivizes teaching and community health.

9.2 Guilds & Social

- Guilds progress via collective Mastery Proofs (guild-wide challenges), fostering group learning.
- Guild halls display a "Resonance Map" showing the guild's collective math mastery — a source of pride and goal-setting.
- Cooperative raid design where different wings require different tiers — a raid *needs* players across tiers, making diversity a strength, not a liability.

9.3 Economy & Crafting

- Mastery Materials are the primary crafting currency, earned through boss defeats and Mastery Proofs.
- Trading is limited to materials and cosmetics, **not power** — preventing pay-to-win and keeping skill central.
- A "Scribe's Market" lets players trade puzzle solutions hints (cosmetic-priced), supporting the learning community.

9.4 Seasonal Events & Live Ops

- **The Grand Tournament:** weekly speed-accuracy leaderboards across tiers (separate leaderboards per tier so Tier-2 players compete against Tier-2, not against Tier-9 veterans).
- **The Discordance:** server-wide world-boss invasions requiring massive cooperative math.
- **Seasonal puzzles:** limited-time proof dungeons with unique cosmetic rewards.

9.5 PvP Constraints

PvP is consensual and tier-matched (no open-world ganking by high-tier players on low-tier ones). Modes focus on competitive problem-solving races rather than raw stat dominance, so skill, not gear, decides outcomes.

10. Engagement & Burnout Risk Assessment

This is treated as a serious product risk, not an afterthought. The central danger is that a math-powered MMO could feel like a **math worksheet with fantasy dressing** — tedious, anxiety-inducing, and exclusionary. Every design decision above exists to prevent that. This section consolidates the risks and mitigations.

10.1 Risk Register

Risk	Description	Severity	Mitigation
Math anxiety	Players feel judged, embarrassed by wrong answers, especially in groups	High	No public failure display; private feedback; assist mode; praise effort; warm-up arenas; mentor support
Worksheet fatigue	Repetitive drilling feels like homework, sapping fun	High	Juicy combat feedback; varied problem formats; rotating challenge types; boss mechanics that demand pattern recognition, not just computation; strong reward cadence
Speed pressure / performance anxiety	Timers induce panic, especially in dyscalculic or anxious players	High	Adaptive timers; generous early windows; visible-but-calm time UI; assist mode; no harsh failure states
Skill-ceiling frustration	Hard gating blocks progression, players feel stuck	High	Multiple paths per tier; cooperative alternatives; spaced practice; mastery measured over time not single attempts; assist mode lowers band
Social toxicity / exclusion	Strong players exclude weaker ones	Medium	Cooperative roles for all skill levels; carry-prevention design; mentor incentives; tiered leaderboards; code of conduct + reporting
Repetitive grind burnout	Repeating same problems for materials	Medium	Daily caps on optimal grind; varied encounter formats; cosmetic alternative reward paths; rested bonuses
Accessibility (dyscalculia, etc.)	Players with learning differences locked out	High	Assist mode (extended timers, multiple choice, reduced complexity, visual/audio cues); full progression achievable in assist mode; no power locked behind speed
Cheating / external tools	Calculators or bots undermine skill integrity	Medium	Server-authoritative validation; per-player randomization; assist-mode matchmaking fairness; design makes external tools slower than playing
Over-session / binge burnout	Players over-grind and burn out	Medium	Daily caps, rested bonuses, soft daily objectives, encouragement of session breaks
Reward misalignment	Rewards don't match effort, feels unrewarding	Medium	Mastery-earned gear (not RNG); visible progression; cosmetic paths; clear feedback on growth

10.2 Core Anti-Burnout Principles

1. **Mastery over speed.** Gear and progression reward demonstrated understanding and consistency, not raw speed. A thoughtful, accurate player outpaces a reckless fast one.
2. **Flow via adaptive difficulty.** The engine keeps every player in their personal flow band — challenged but never crushed.
3. **Failure is informative, never humiliating.** Wrong answers reduce effectiveness but never exclude or embarrass.
4. **Multiple paths forward.** If a player struggles with one operation, alternate routes, cooperative roles, and assist modes keep them progressing.

5. **Cooperation, not competition, as default.** The social fabric rewards helping, teaching, and contributing at any level.
6. **Variety in challenge format.** Combat fluency, puzzle understanding, pattern recognition, and proof-dungeons rotate to prevent monotony.
7. **Rest and caps.** Daily caps on optimal grind and rested bonuses prevent binge-burnout and keep the game a sustainable long-term habit.
8. **Intrinsic + extrinsic reward alignment.** Players genuinely improve at math (stealth learning), which is intrinsically satisfying, while earning tangible gear and social status extrinsically.
9. **Full accessibility paths.** Assist mode is a first-class progression path, not a second-class workaround. No power is locked behind speed or a specific operation strength.
10. **"Learn by playing" tutorials.** Onboarding teaches the math *through* the combat fantasy, so players grow without ever feeling they're being schooled.

10.3 Accessibility & Assist Mode (Detail)

Assist Mode is a first-class, non-stigmatized option:

- Extended answer timers (configurable multiplier).
- Multiple-choice answers (reduces motor + recall cost, keeps cognitive engagement).
- Reduced problem complexity (one sub-tier lower within the tier's band).
- Visual + audio cues (e.g., color-coded operation signs, optional spoken problems).
- Full progression and gear access — assist-mode players reach all content, just with a flagged matchmaking tier for fairness.
- Designed with input from dyscalculia and accessibility consultants during development.

10.4 Engagement Hooks (Why Players Stay)

- **Intrinsic mastery:** the genuine satisfaction of getting measurably better at math, made visible through the world opening and gear unlocking.
- **Extrinsic rewards:** epic gear, cosmetics, titles, mounts, leaderboards.
- **Social belonging:** guilds, raids, mentor relationships, cooperative stakes.
- **Stealth learning:** players improve a real-world skill, which is its own deep reward — a rare and powerful retention driver for the target audience.
- **Living world:** seasonal events, world bosses, and server-wide Discordance invasions keep the world feeling alive.

10.5 Success Metrics & KPIs

KPI	Target / signal
First-boss completion rate	>85% of new players defeat first world boss
Retry rate after first failure	Low; failure should invite retry, not quit
Tier-to-tier drop-off	<15% per tier; spikes indicate a difficulty or fatigue problem
Avg problem response time	Trending down within a tier = skill growth
Wrong-answer frustration quits	<5% of sessions end within 30s of a wrong answer
Assist-mode adoption & progression	Assist players reach endgame at similar rates
Daily challenge completion	High and stable = healthy engagement
Session length distribution	Healthy spread, no spike in 4h+ binge sessions
Burnout indicators (declining login streak then churn)	Monitored; rest bonuses triggered automatically
Guild participation	>60% of active players in a guild
Player-reported math-skill improvement	Surveyed; a core brand promise and retention signal

10.6 Continuous Burnout Monitoring

The live-ops team monitors aggregated engagement signals (not individual performance) to detect zone-level frustration spikes, then tunes adaptive bands, timers, and reward cadence in hotfixes. Player feedback channels (in-game, community) feed a standing "Mastery & Motivation" review each season.

11. Monetization (Summary)

- **Free-to-play**, all power-earning is skill-based and non-monetized.
- **Cosmetics only**: skins, mounts, titles, emotes — never power, never problem-solving aids that would constitute pay-to-win.
- **Optional convenience** (e.g., extra daily practice tokens) — capped so it cannot bypass mastery.
- **No loot boxes for power**. Cosmetic crates are direct-purchase or earnable.
- The commercial model explicitly protects the skill-integrity promise, since the entire game collapses if power can be bought.

12. Technical & Production Notes (Summary)

- Server-authoritative math validation; per-player problem randomization.
- Adaptive difficulty engine: per-player, per-operation response-time + accuracy model.
- Cross-platform input: keyboard + number pad (PC/Mac), touch numeric pad (tablet).
- Accessibility-first input layer (voice, larger hit targets, color/contrast).

- Stylized low-poly art direction keeps performance high and readability strong (effects never obscure problems).
- Designed for modular content authoring: each Dominion is a self-contained content pack (art + problem set + boss + gear), enabling a sustainable live-ops cadence.

13. Open Questions & Next Steps

1. Finalize the exact problem-count-per-minute target per tier (playtesting required).
2. Validate adaptive-difficulty bands with a representative playtest cohort, including dyscalculic players.
3. Prototype the 90-second boss phase and measure perceived-fun vs. perceived-effort.
4. Define the Mastery Proof pass/fail thresholds per tier.
5. Art-direction pass to ensure effects never obscure problem text (a recurring risk in stylized combat).

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